

SUGESTÕES DE RESOLUÇÃO DE INCIDENTES ATRAVÉS DO USO DE LLM'S

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SOBRE A EMPRESA



**Critical
Techworks**

A BMW GROUP &
CRITICAL SOFTWARE
COMPANY



Connected Car



Autonomous
Driving



Cloud Solutions
& Cyber Security



BMW Vehicle
Development



Marketing, Sales
& After Sales



Production
and Logistics



Data Intelligence

CONTEXTO



Gestão de incidentes



Projetos críticos



Devops

O QUE É UM INCIDENTE

Incident

Number

INC0001

Channel

Standard-UI

Caller *

Technical identity for

State

New

Originator group

Impact

3 - Low

Service

Urgency

2 - Medium

Service offering *

Priority

4 - Low

Configuration item

Assignment group *

Notes

Watch list

Work notes list

Additional comments (Customer visible)

Work notes

Incident

Short description *

CloudWatch test environment test-service-api-ecs-task-stopped-unexpectedly ECS Task fail

Description *

CloudWatch test environment test-service-api-ecs-task-stopped-unexpectedly
ECS Task failed with stopCode ServiceSchedulerInitiated or EssentialContainerExited
Warning alert [Alert00000XXXXXX] .
Created on [AccountX].
Metric name is [test-service-api-ecs-task-stopped-unexpectedly-alarm] of type [] from data
source [AWS].

Total impacted services by the CI: [Product Y (Monitoring)] is 1.

Resolution Information

Resolution code

No resolution provided

Resolved by

—

Resolved

—

Resolution notes

This incident was due to high use of memory from 1 instance. Restarting all instances
temporarily solved the issue.
Later the team will look with closer attention to the Root Cause.

PROBLEMA



Tempo de resposta

Processo de análise e mitigação



Sobrecarga

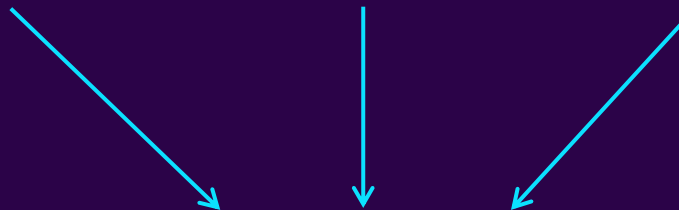
Duplicados ou demais



Experiência / Ambiente

Elementos inexperientes

Momentos de tensão/confusão



Custos e Disponibilidade



Serviços não disponíveis

Clientes insatisfeitos

OBJETIVOS



Resumos de incidentes

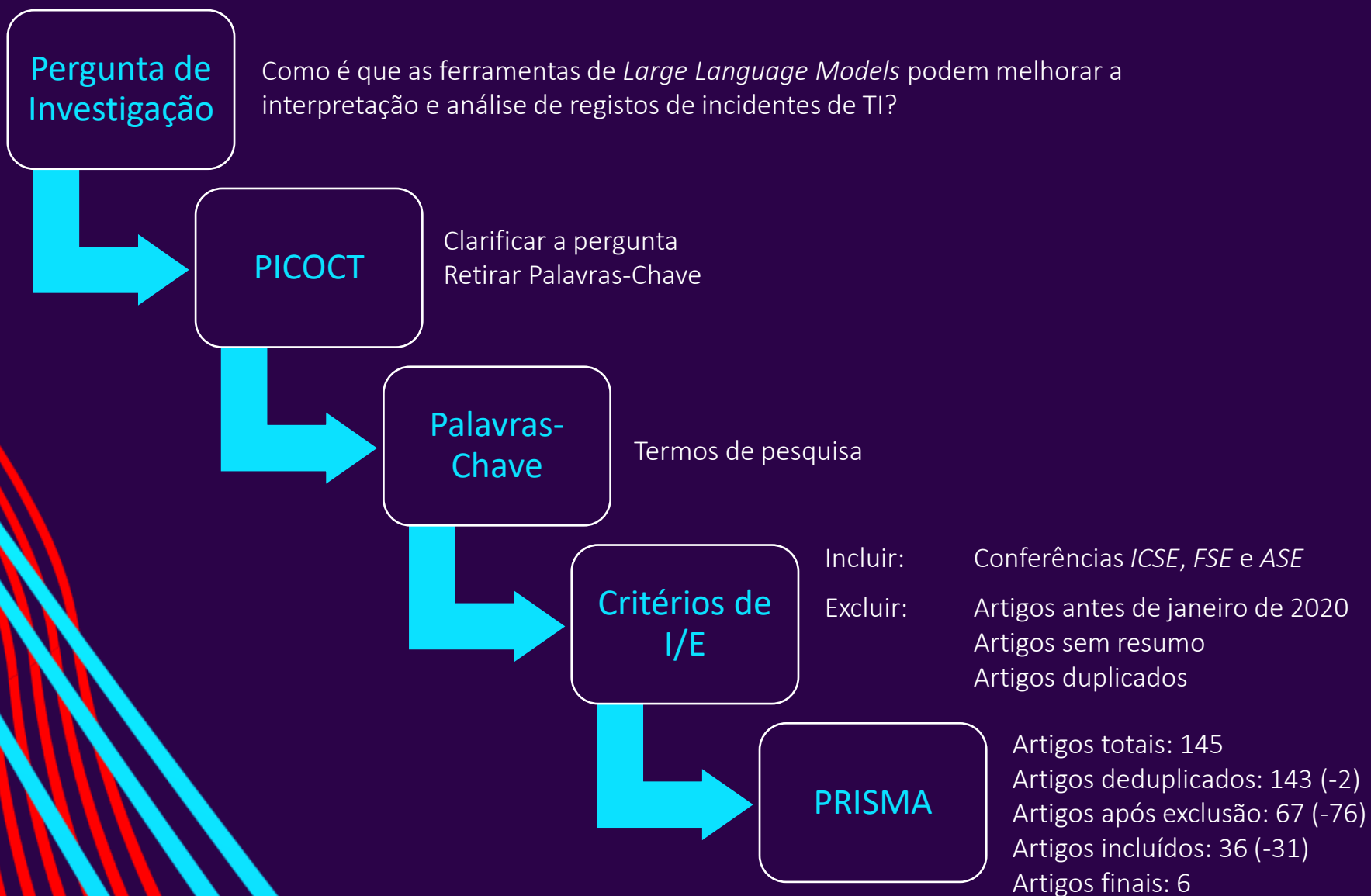


Análise de registos



Sugestões de mitigação

METODOLOGIA DE INVESTIGAÇÃO



ARTIGOS REVISTOS



Xpert: Empowering Incident Management with Query Recommendations via Large Language Models



X-Lifecycle Learning for Cloud Incident Management using LLMs



Exploring LLM-Based Agents for Root Cause Analysis



How to mitigate the incident? an effective troubleshooting guide recommendation technique for online service systems



Identifying linked incidents in large-scale online service systems



Dynamic Graph Neural Networks-Based Alert Link Prediction for Online Service Systems

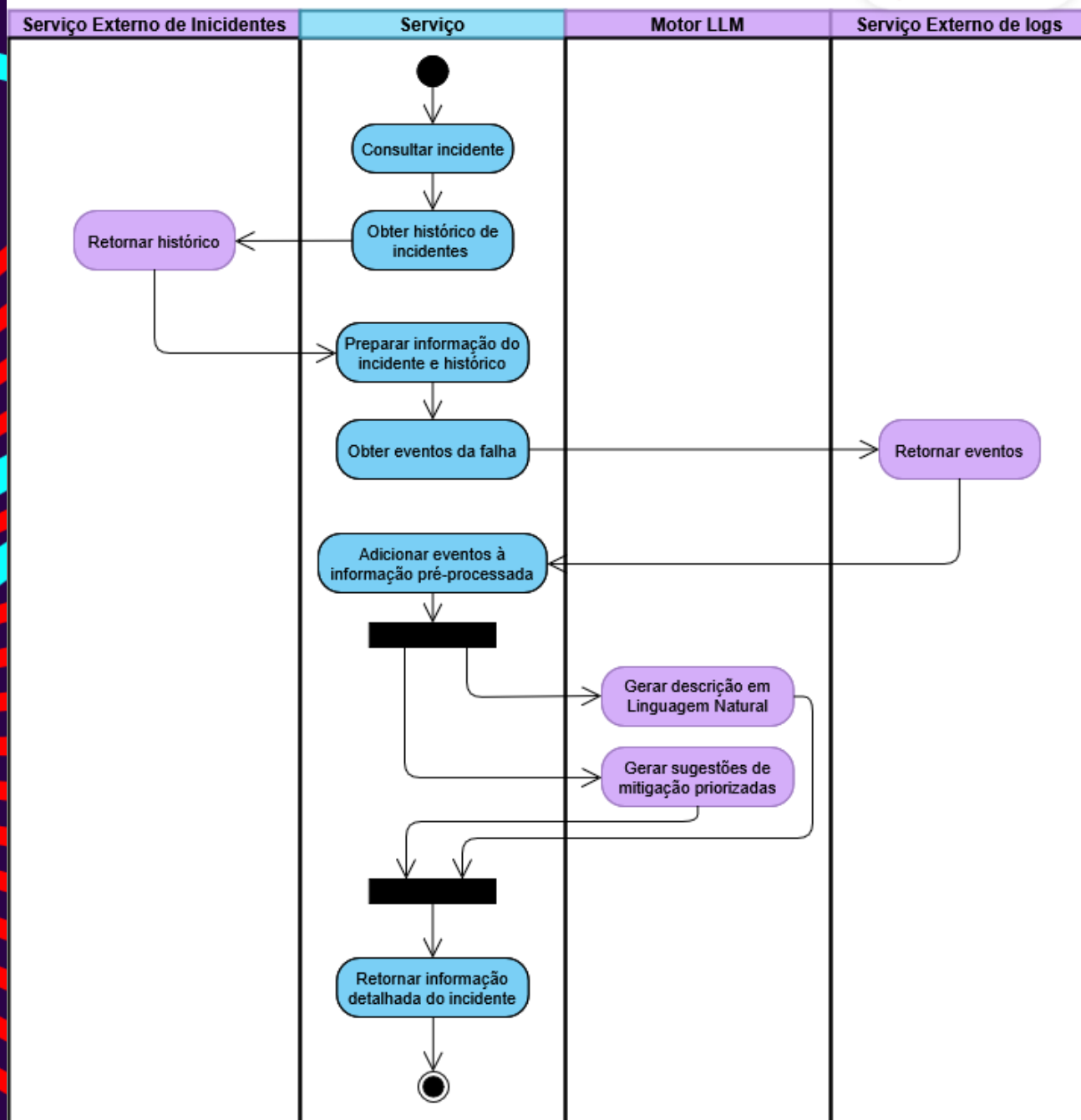
SOLUÇÕES EXISTENTES

AWS

NEW RELIC

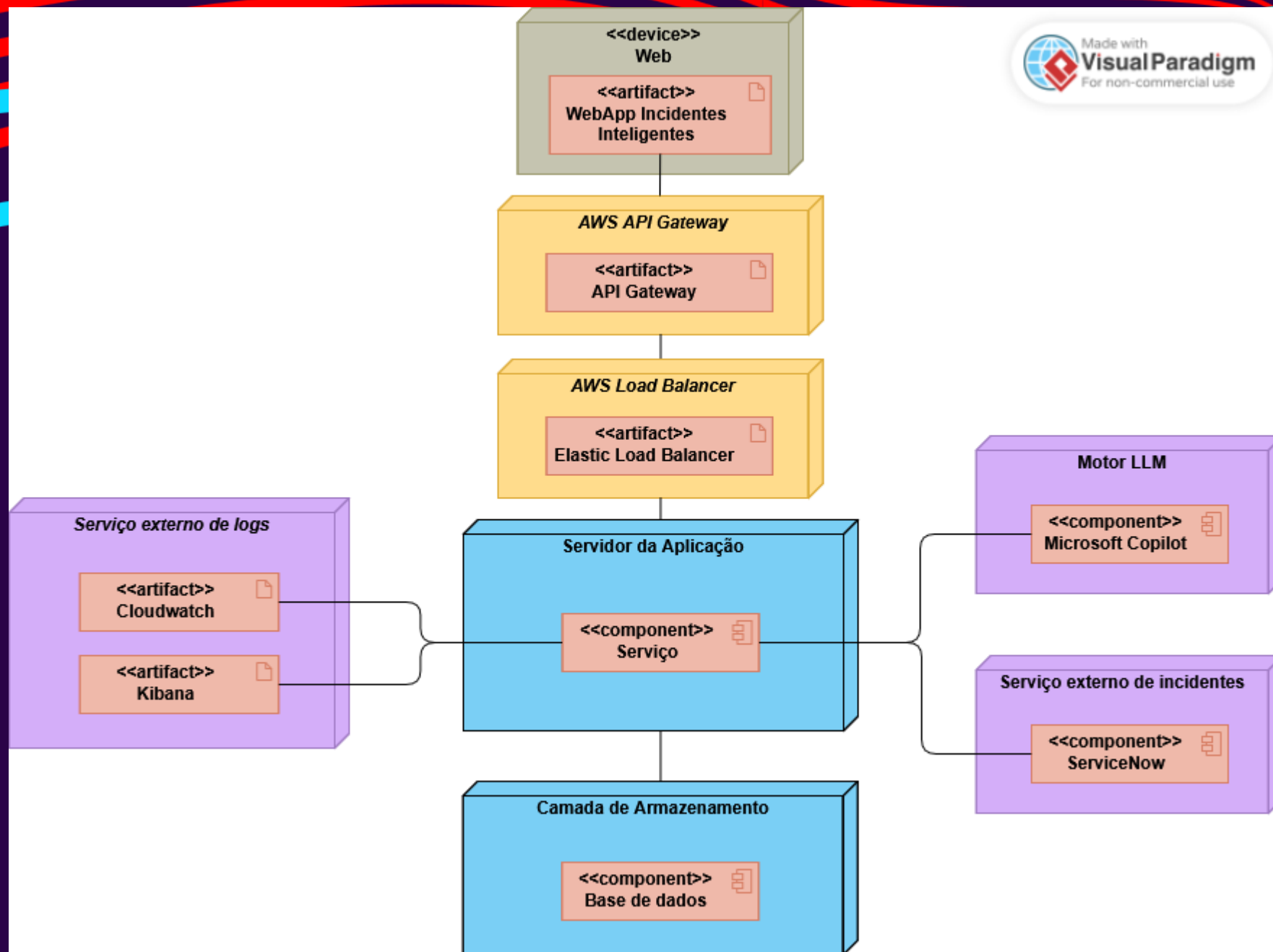
ATLASSIAN

ELASTIC



METODOLOGIA PROPOSTA

Diagrama de Atividade



METODOLOGIA PROPOSTA

Diagrama de Implantação

PRESSUPOSTOS

P1 | O conjunto de dados será entregue por uma das equipas da empresa num formato passível de utilização

Plano B | Conjunto de dados sobre incidentes de plataformas reconhecidas

P2 | O motor *LLM* a utilizar será disponibilizado pela empresa de forma a não comprometer dados

Plano B | Desenvolver um *Small Language Model (SLM)* de raiz

Plano C | Conjunto de dados sobre incidentes de plataformas reconhecidas

RESTRIÇÕES

Os dados presentes no *dataset* fornecido pela empresa terão de ser parcialmente/totalmente anonimizados.

OBRIGADO